



Production Information Systems

Chapter 4 : Structured Analysis and Functional Architecture Design

ISE Department
Year 4/ Level 7

Instructor : Dr. Jomana Bashatah



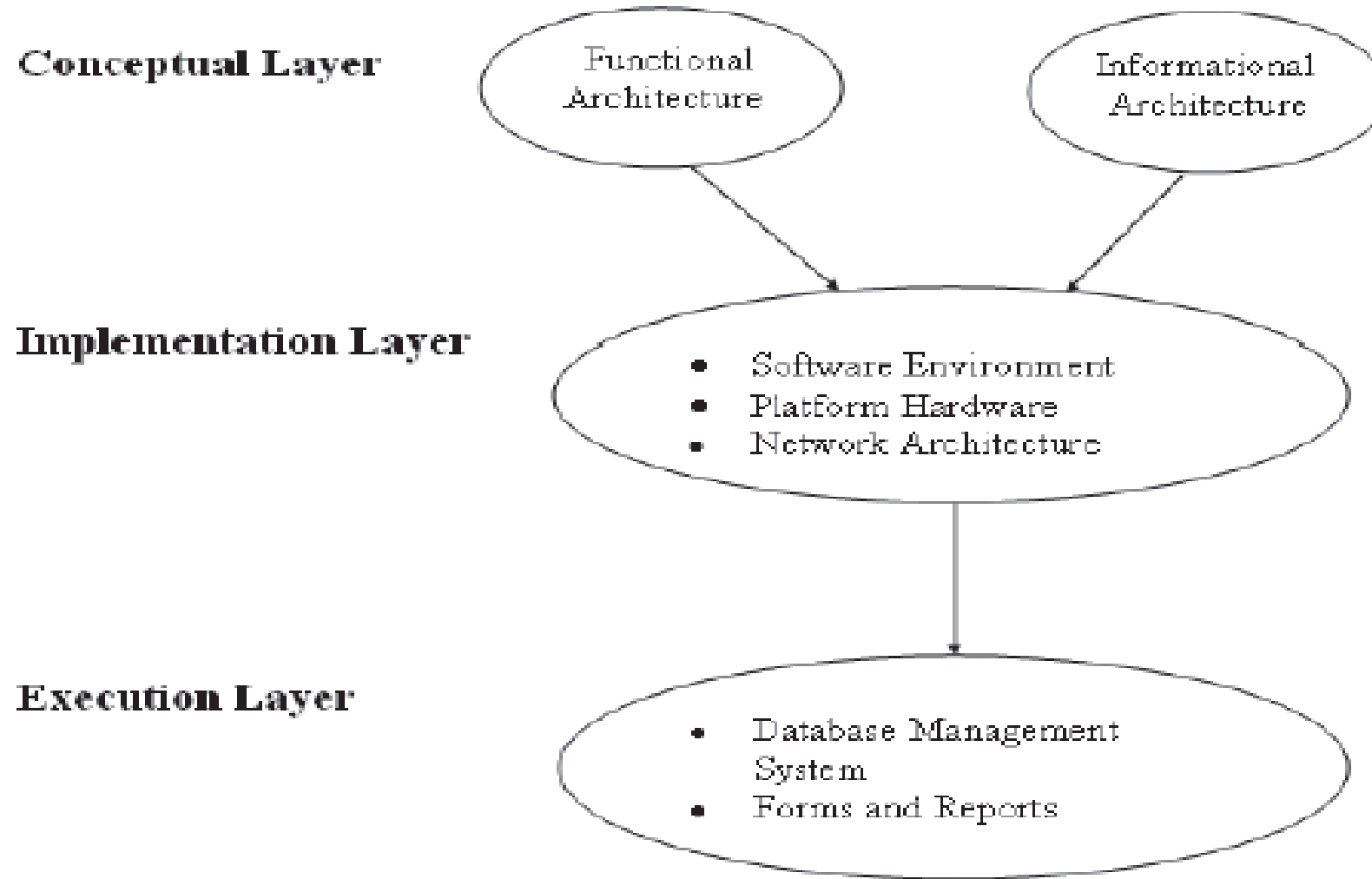
Outline

1. Introduction
2. Functional Architecture and Business Process Redesign
3. IDEF0 Methodology Modeling
4. Data Flow Diagram (DFD)

1.Introduction

- In this chapter, we focus on the first step in the design of an information system for an industrial enterprise. The design proceeds from a definition of a **business model** of the enterprise.
- ***The business model*** is a description of the functions of the business, the data requirements, and the interactions between the functions and data requirements.
- The term ***functional or activity architecture*** is often used to describe a conceptual model of the activities that operate the business and the relationships between those activities.
- A related ***conceptual model*** of the firm is the ***informational or data architecture***. This is a model of the information requirements needed to perform the functions of the business.

1.Introduction



Layers of the information system design process.

1.Introduction

- **The *conceptual layer*** consists of the logical design of function and data requirements.
- When a conceptual design is complete, the next step is to implement the ***blueprint*** in hardware and software.
- **The *implementation layer*** requires the selection of a database management system, hardware platforms, and a communication medium. At this level, specific commitments are made that place limitations on the actual realization of the concept models.
- **At the *execution layer***, the concept models are coded in software in terms of forms and reports that are to interface with the individuals performing the functions defined in the functional architecture.

In this chapter, we focus on the conceptual layer and, in particular, the functional architecture design methodology.

2.Functional Architecture and Business Process Redesign

- The functional architecture conceptual design has a lot in common with the practice called ***business process redesign (BPR)***.
- **BPR** is a term given to the study of the existing functionality and information used by the enterprise with the intention of redesigning the system to meet the same business objectives at higher performance or lower costs.
- As in BPR, the process of designing a functional model follows a stepwise procedure :
 - The first step is to study the existing enterprise and to deduce the business rules that govern its functions. The resulting model of the enterprise as it currently exists is referred to as the **as is** model.

2.Functional Architecture and Business Process Redesign

- With an understanding of the business rules governing the functions, the objective is to eliminate redundancy and functions that are not required to meet the business objectives.

The resulting model of the enterprise is called the **to be** model.

Business process redesign focuses on the *process*, which is a set of logically related functions that are organized to achieve a goal. These processes have customers that they are trying to service. Those customers may be outside the enterprise, as in the case of consumers of the final product, or they may be within the enterprise, as in the case of internal consumers of information or services generated by the process.

As an example of a process in manufacturing, consider the replenishing of raw materials. It may begin in production planning with a requisition, flow through purchasing, go to an outside vendor, returning materials to receiving.

2.Functional Architecture and Business Process Redesign

- There are two methodologies that we will examine for designing a functional architecture. Both of these methodologies are known as “**structured analysis**” techniques.
- When an analyst approaches an information system project, he or she is often unfamiliar with the “**business context**” wherein the system will be implemented. The business context includes the functions performed by employees and the relevant data that are used to perform those functions. There is a learning process that the analyst must go through during which the functions are documented. The procedures and methods used in the documentation process are called “**structured analysis**” methods.
- One application of structured analysis is the use of **data flow diagrams (DFD)**, which have been promoted in industry by Yourdon et al. (1979). This method is widely used by information system professionals in all industries. An alternative method is **structured analysis and design technique (SADT)** (Marca et al., 1988), which has been adapted for manufacturing enterprises under the name **integrated computer-aided manufacturing definition 0 (IDEF0)**.

3. IDEF0 Methodology Modeling

IDEF0 is a modeling methodology for designing and documenting hierarchic, layered, modular systems.

It is considered to be primarily relevant to manufacturing systems.

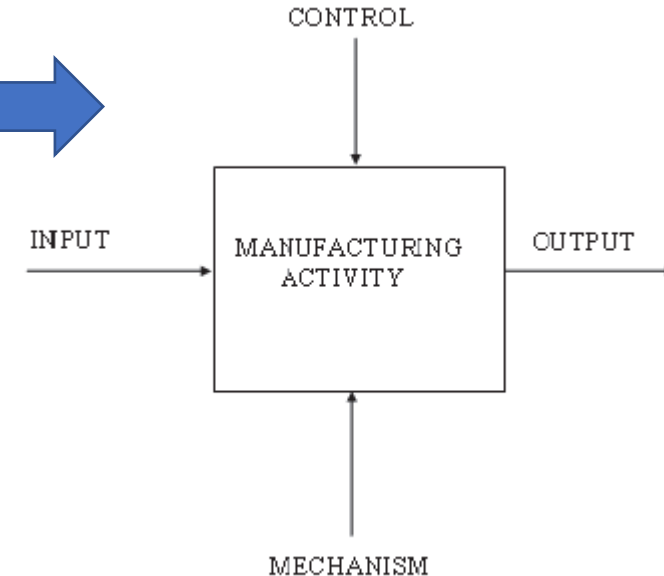
The **activity box** is used to describe a function being performed in the enterprise.

Typically, this is either a material conversion function, such as machining a part, or an information conversion function, such as processing a requisition for ordering materials. The same generic activity box primitive can model any function.

The building blocks of IDEF0



*As an example of a **material input**, it could be a workpiece to be machined. As an example of an **information input**, it could be the information on a requisition to be transformed into a **purchase order (output)**. The information conversion to process a requisition into a purchase order could involve a **purchasing agent** as the **mechanism**. The machining activity may require a **numerical control parts program** if the machining is automatic or a **set of mechanical drawings** if the machining is done semi-automatically (**control**).*



IDEF0 activity box and connecting arrows.

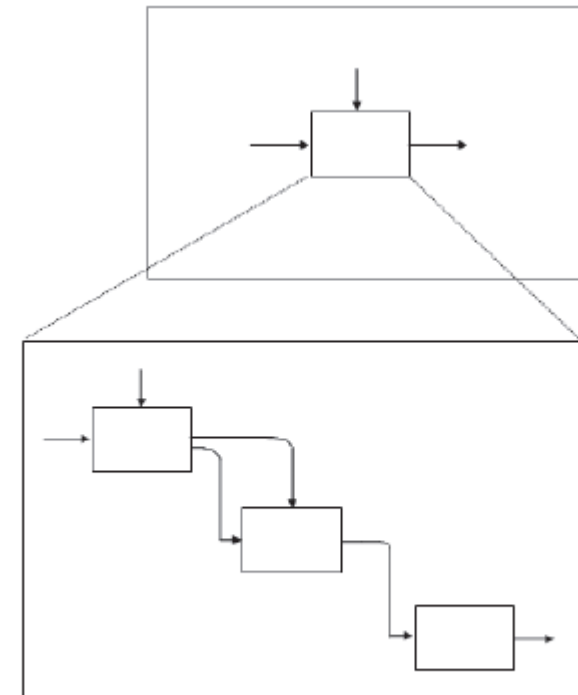
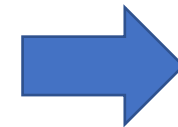
3. IDEF0 Methodology Modeling

- **Inputs** are those things that are transformed by the function
- **Outputs** are the result of the transformation process provided by the activity
- **Mechanisms** are the means by which a function is realized.
- **A control** is a condition or set of conditions that guide or constrain the performance of the activity.
- An **input** is transformed into an **output** by an *activity* (function) performed by a **mechanism** and governed by a **control**. The specific activity, inputs, outputs, mechanisms, and controls are defined by the situation being modeled.
- Activities represent actions being performed and are labeled with verb phrases. Inputs, outputs, mechanisms, and controls represent things and are labeled with noun phrases.

3. IDEF0 Methodology Modeling

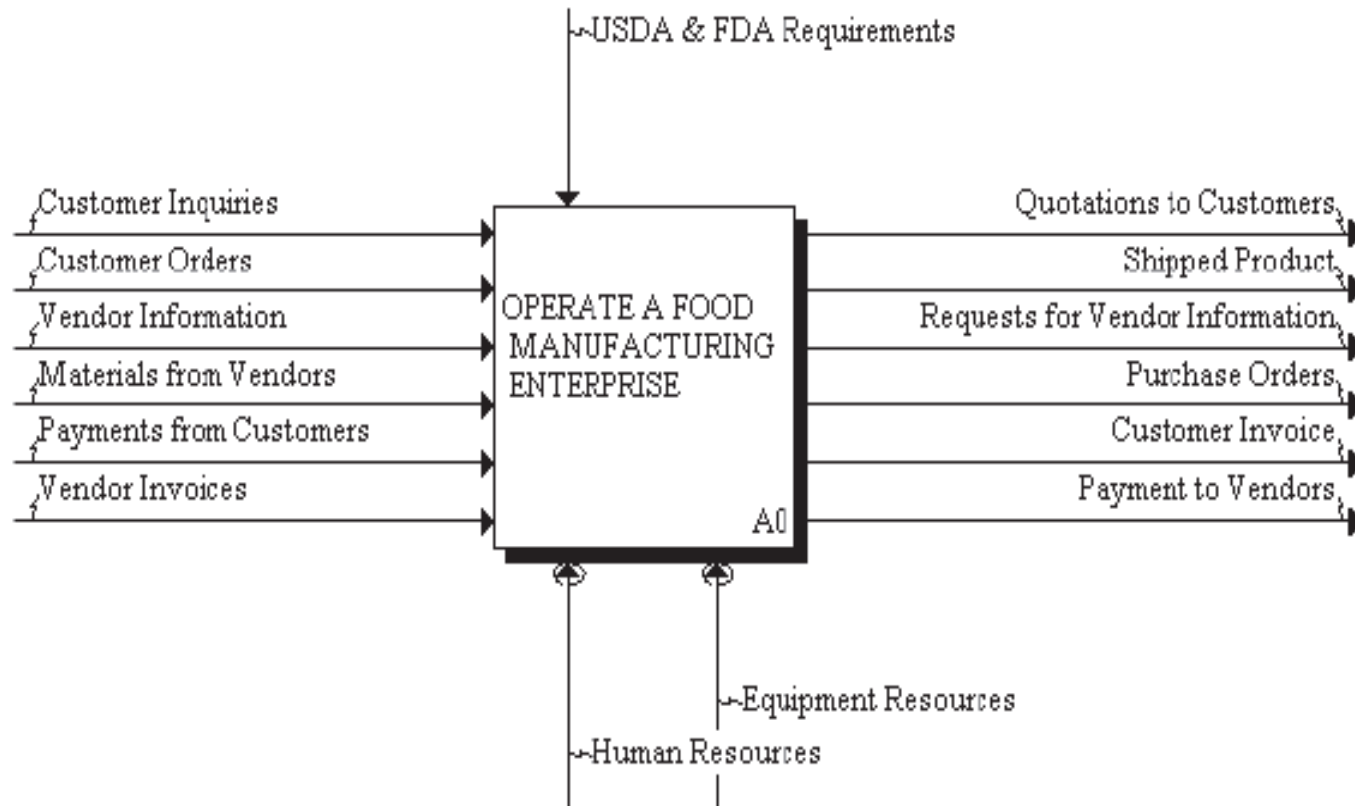
IDEF0 Hierarchic Decomposition

- IDEF0 is a top-down modeling approach
- The first layer is a single activity box that describes the overall function of the enterprise, organization, or process within the enterprise that is the subject of the model.
- This overall activity is then decomposed into its major subactivities at the second layer. Each of these subactivities are further decomposed at the next layer, and so forth. Functions are related to each other by their material flows and information flows.
For example, the output material or information of one activity may provide the input to another activity.



3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example



Top-level view of the enterprise: Node A0.

Suppose an analyst were given the problem of developing an integrated model of an entire manufacturing enterprise. The place to begin would be with a definition of the overall activity of the enterprise.

This Figure is a high-level view of the enterprise and its interactions with the outside world.

The inputs and outputs of interest at this level are the important input-output relationships at the boundary of the enterprise and the outside world that undergo the conversion process of the enterprise activity.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

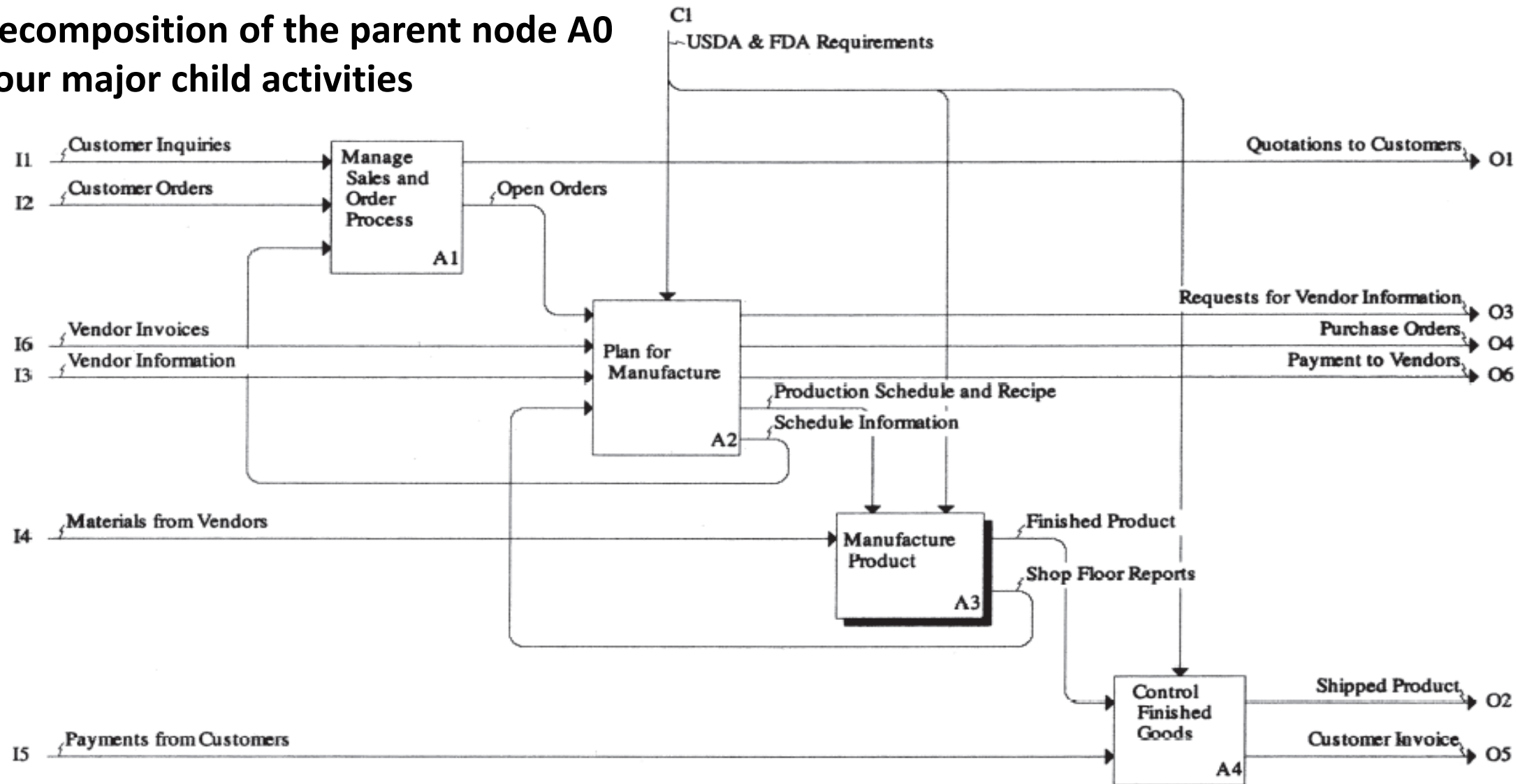
To a large extent, the operation of a manufacturing enterprise can be viewed as three interrelated processes:

- First is the **physical flow of materials**. At node A0, this is represented by the input *materials from vendors*, which is transformed into the output *shipped product*. This physical conversion process is what we normally think of as the manufacturing process.
- The second process is **the information flow**. Internal to the organization, it is the information flow that leads to decisions that affect the physical flow system. At node A0, such items as *vendor information* and *quotation to customers* are typical of information flows.
- Finally, there is **the financial flow system**, which results in the collection and disbursement of wealth. Payments from customers and payments to vendors are obvious examples.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

The decomposition of the parent node A0 into four major child activities



3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

The important factor is that, when taken as a set, these A1 to A4 activities are a complete representation of the components of activity A0 :

- A0 — Operate a Food Manufacturing Enterprise
- A1 — Manage Sales and Orders Process
- A2 — Plan for Manufacture
- A3 — Manufacture Product
- A4 — Control Finished Goods

The higher level activity from which a decomposition occurs is called the ***parent activity***, and the decomposed activities are called ***child activities***. Inputs and outputs of the parent activity are also inputs and outputs of the child activity

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

The continuation of the process of decomposition will be illustrated on node **A3**,
Manufacture Product

First, let's define the node numbering convention. The convention uses the parent node number with an extension. Thus, we have **A31**: *Control Incoming Materials*, **A32**: *Control Stored Materials*, and **A33**: *Control Production Processes*.

At this point the indented diagram structure is as follows:

A0 — Operate a Food Manufacturing Enterprise

A1 — Manage Sales and Order Process

A2 — Plan for Manufacture

A3 — Manufacture Product

A31 — Control Incoming Materials

A32 — Control Stored Materials

A33 — Control Production Processes

A4 — Control Finished Goods

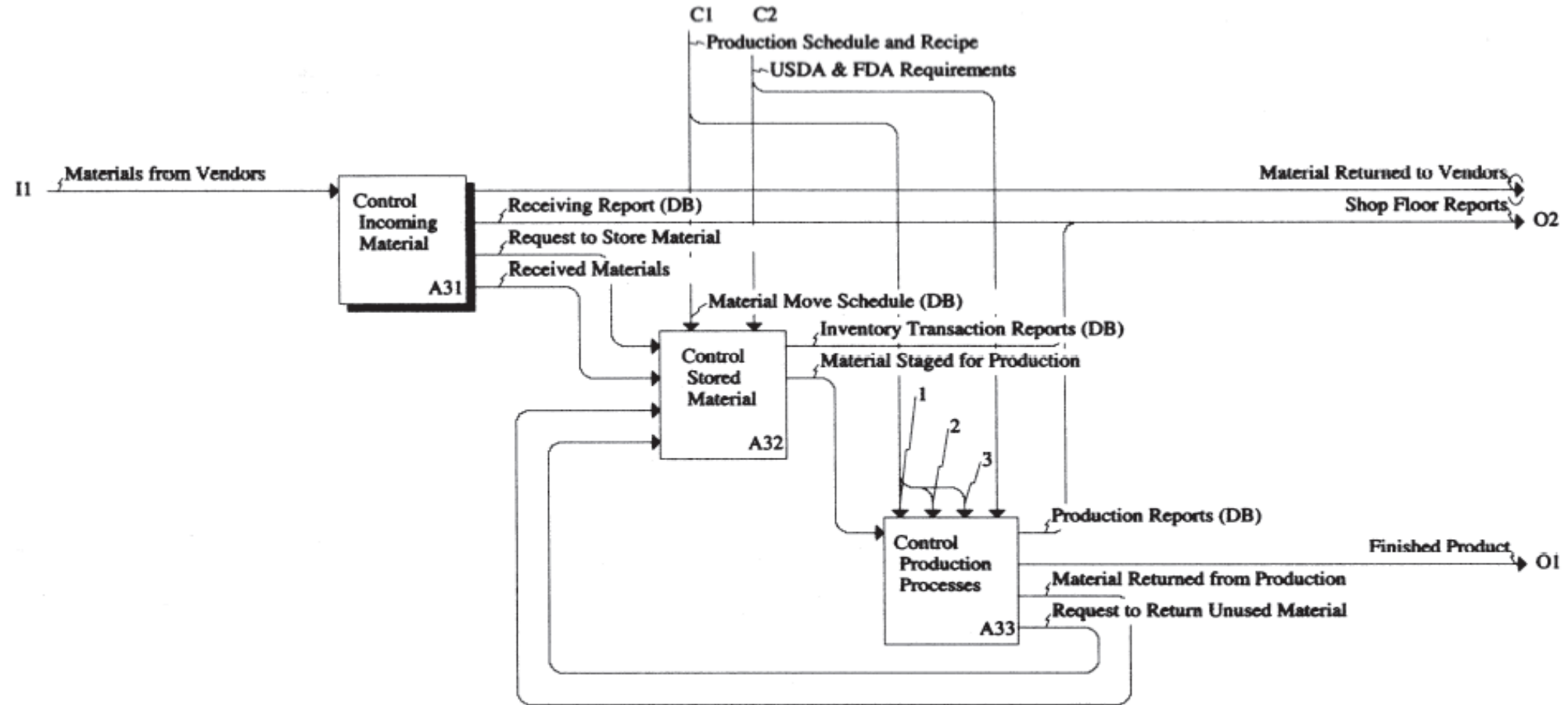
3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

- An additional modeling concept, called ***bundling***, is introduced in this diagram.
- The activity node A2 provides a control for activity node A3 called *production schedule and recipe*. The production schedule allocates products to be produced to the equipment (production lines) by time periods. The recipe is composed of many things. It includes the **steps in the production process, the materials** or ingredients that are used at each step to make the product, and **the critical operating parameters** of the production line (i.e., temperatures and time settings for cooking and sterilization). All are subsumed under the concept of a “**recipe**.”
- In Figure below we have reached a level of decomposition where it is time to make these controls more explicit.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example



Decomposition of node A3.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

- The figure introduces the concept of ***tunneling***. An example is shown by the output of activity A31, labeled “**Material Returned to Vendors.**” Note the tunnel on the arrowhead of the arc. A tunnel arrow is used as a convenience. It can represent: (1) an external arrow that did not appear in the parent diagram (i.e., it has a hidden source) or (2) an arrow that goes to another activity but does not appear explicitly on the destination activity (i.e., a hidden destination).
- Tunneling is used when it is not convenient to show all inputs, outputs, controls, or mechanisms at every level of the hierarchy.
- In completing the decomposition down to the elemental level, we will focus on node A31,
- Control Incoming Material. The elemental level is the most detailed level of analysis of functions.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example

Node A31 has been decomposed into three child activities. At this point, the indented list of activities is as follows:

A0 — Operate a Food Manufacturing Enterprise

A1 — Manage Sales and Order Process

A2 — Plan for Manufacture

A3 — Manufacture Product

A31 — Control Incoming Materials

A311 — Confirm Validity of Shipment

A312 — Inspect Condition of Materials

A313 — Receive Materials

A32 — Control Stored Material

A33 — Control Production Processes

A4 — Control Finished Goods

3. IDEF0 Methodology Modeling

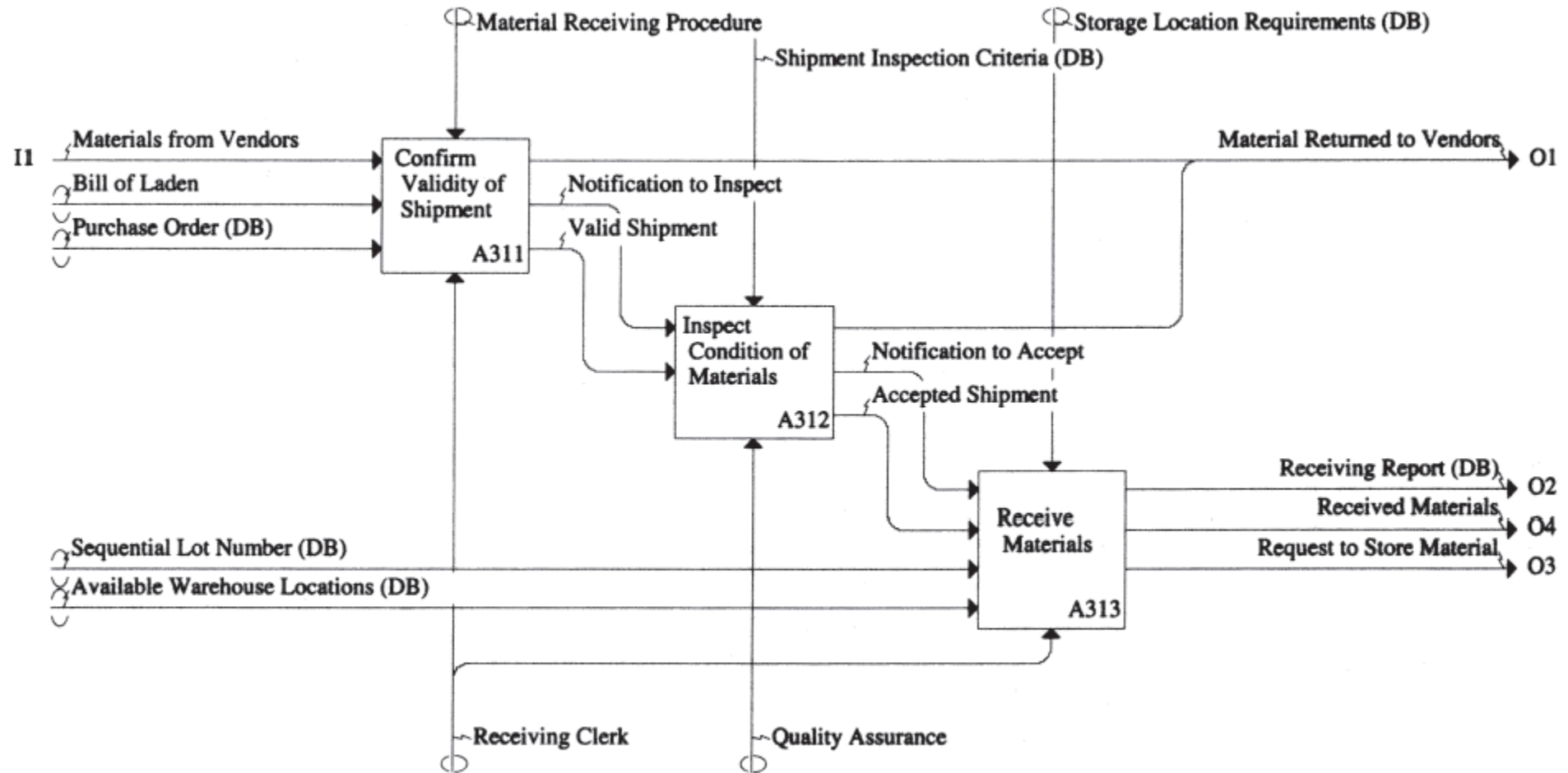
IDEF0 Hierarchic Decomposition : Example

This diagram and narrative are developed with those individuals within the enterprise who are actors in the process. For the reader to understand the details of activities A311 to A313, the narrative as developed by the analyst during the modeling process will be described as follows:

Node A311: Confirm validity of shipment. When a delivery truck arrives at the receiving dock, the receiving clerk is responsible for clearing the material for unloading. The clerk first compares the paperwork that comes with the shipment with the enterprise's purchase order. Therefore, two required information inputs to this function are the bill of lading and the corresponding purchase order. If the material on the shipment bill corresponds to the purchase order, the shipment is accepted. If no such purchase order exists, the shipment is refused. This procedure is defined as the "material receiving procedure." The steps of this procedure, as defined by the management of the enterprise, control the manner in which the function is performed.

3. IDEF0 Methodology Modeling

IDEF0 Hierarchic Decomposition : Example



Decomposition of node A31.

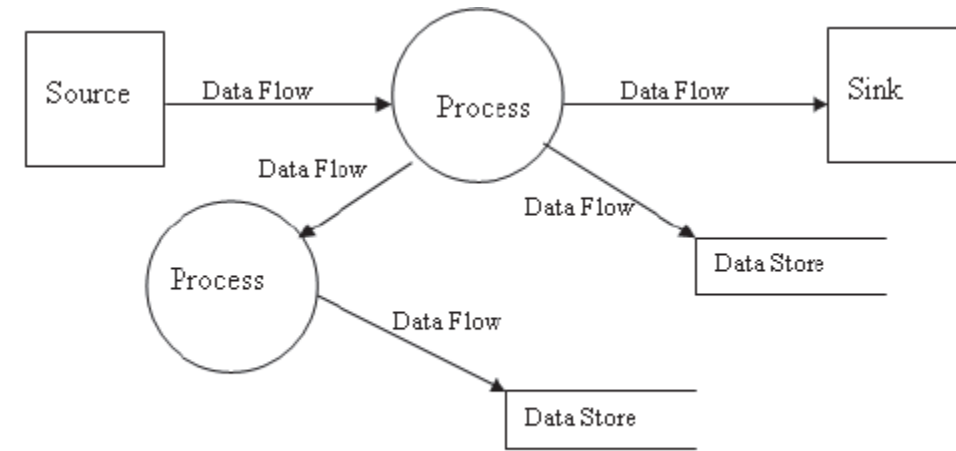
4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

There is an alternative to IDEF0 that is widely used in all industries, both manufacturing and service industries, called ***data flow analysis (DFA)***. Data flow analysis also models functions and the information that flows among functions. The focus of DFA is on the movement of data among those functions. It explicitly includes data repositories, called “data stores.” Unlike IDEF0, it is not concerned with material flows or with the mechanisms that perform each activity.

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

DFA modeling

- Data flow diagrams are constructed using four symbols
- The ***process***, which is analogous to an IDEF0 activity, is shown as a circle. The process represents the people and procedures that transform data.
- In DFA, the emphasis is on how the data are used and transformed.
- Each process has data entering it and exiting it.
- An arrow is used to indicate ***data flow***.
- In DFA, “data” is a general concept. It can represent data sent to a computer file or information given from one process to another process by word of mouth, via e-mail, or by a telephone call. Note that arrows are not used to indicate physical flows of materials, as in IDEF0. The emphasis is on the flow of data.



Generic data flow diagram.

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

DFA modeling primitives

- The ***data store*** is represented by a rectangle with an open end. The data store is a place where data are preserved as a record. This could be a computer file or a paper filing cabinet. When the analyst finds that the process results in the storage of a document or a record, a data flow arrow goes from the process to a data store.
- As in the case of IDEF0, the subject of the DFA is defined with reference to a specific system with boundaries. At the boundaries, the system interacts with outside people, processes, and organizations.
- These are indicated by a box, which represents ***sources and sinks*** for data. Sources are entities outside the system that provide data input to the system. They are usually the entities that trigger events in the system. Sinks are entities outside the system that receive data. The same entity may be both a source and a sink if it both sends data to and receives data from the system.

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

Hierarchic Decomposition in DFA

Data flow diagrams follow the same principles of hierarchic decomposition as IDEF0. The highest level diagram is called the ***context diagram***.

The context diagram includes the overall process and the sources and sinks that interact with the overall process.

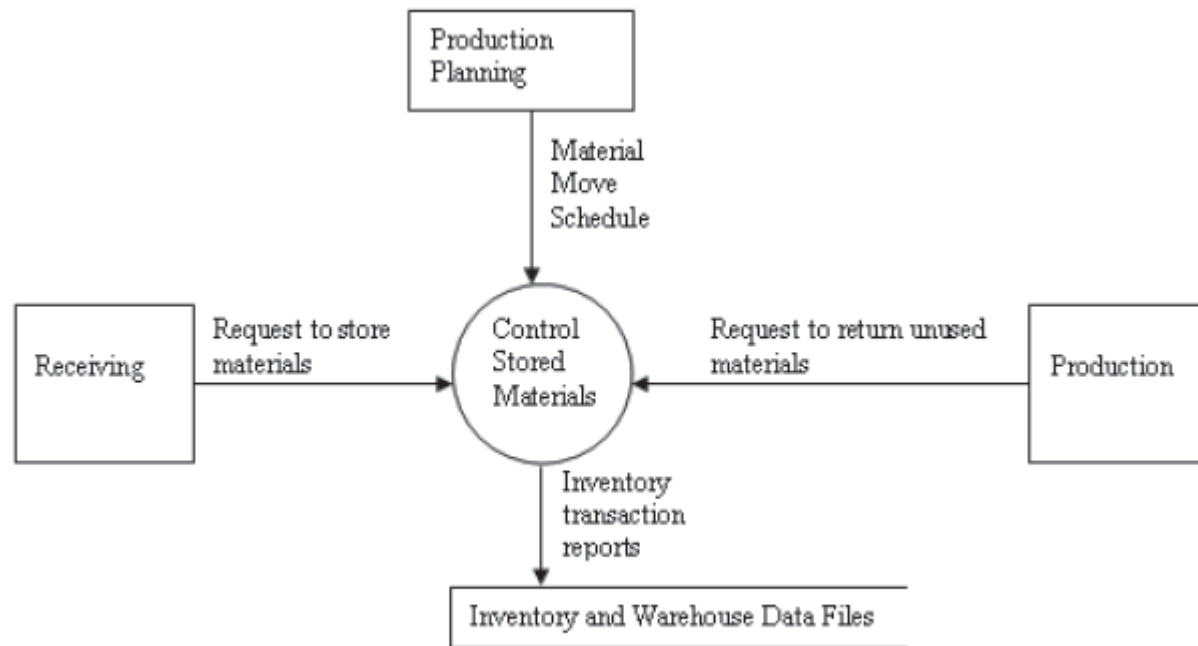
The context diagram is decomposed into a first-level diagram that shows more details of the process and data flow. The boundary relationships of the context level are maintained at each successive level of decomposition. Therefore, data flows from sources and to sinks that appear at the context level also appear at the first level of decomposition.

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

Hierarchic Decomposition in DFA

To demonstrate the use of data flow diagrams, assume that the system of interest has been identified as all the activities involved in node “*Control Stored Materials*”.

A context diagram in the data flow diagram format would appear as shown here:



Data flow diagram context diagram for control stored materials.

There are three source entities at the boundary of the system:

- receiving,
- production planning,
- production

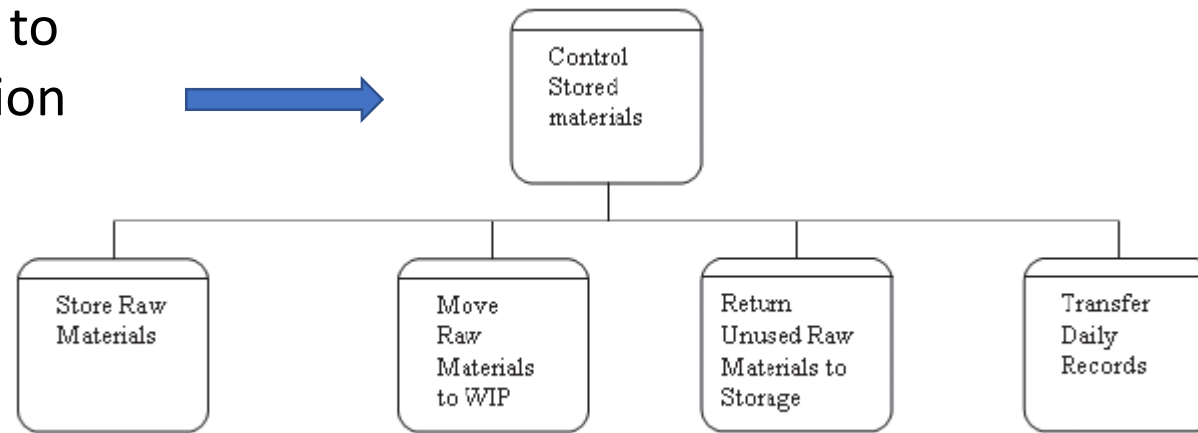
4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

Decomposition of context DFD

In IDEF0 methodology, the overview of the decomposition of the highest level node is maintained by using an indented list. In a similar manner, the overall structure of the data flow diagram hierarchy is often shown in a ***process hierarchy chart***.

The process hierarchy chart is a series of block diagrams that show the hierarchic relationship among processes that are documented in the data flow diagrams.

an overall guide to
the decomposition
of processes



Process hierarchy chart.

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

Steps :

- 1- Create a list of activities
- 2- Construct Context Level DFD (identifies external entities and processes)
- 3- Construct Level 0 DFD (identifies manageable subprocess)
- 4- Construct Level 1/ n DFD (identifies actual data flows and data stores)
- 5- Check against rules of DFD

4. Data Flow Diagrams (DFD): an alternative structured analysis methodology

External Entity  **Noun**

Data Flow  **Names of Data**

Process  **Verb phrase**

- A system name
- A subsystem name

Data Store  **Noun**

4. Data Flow Diagrams (DFD)

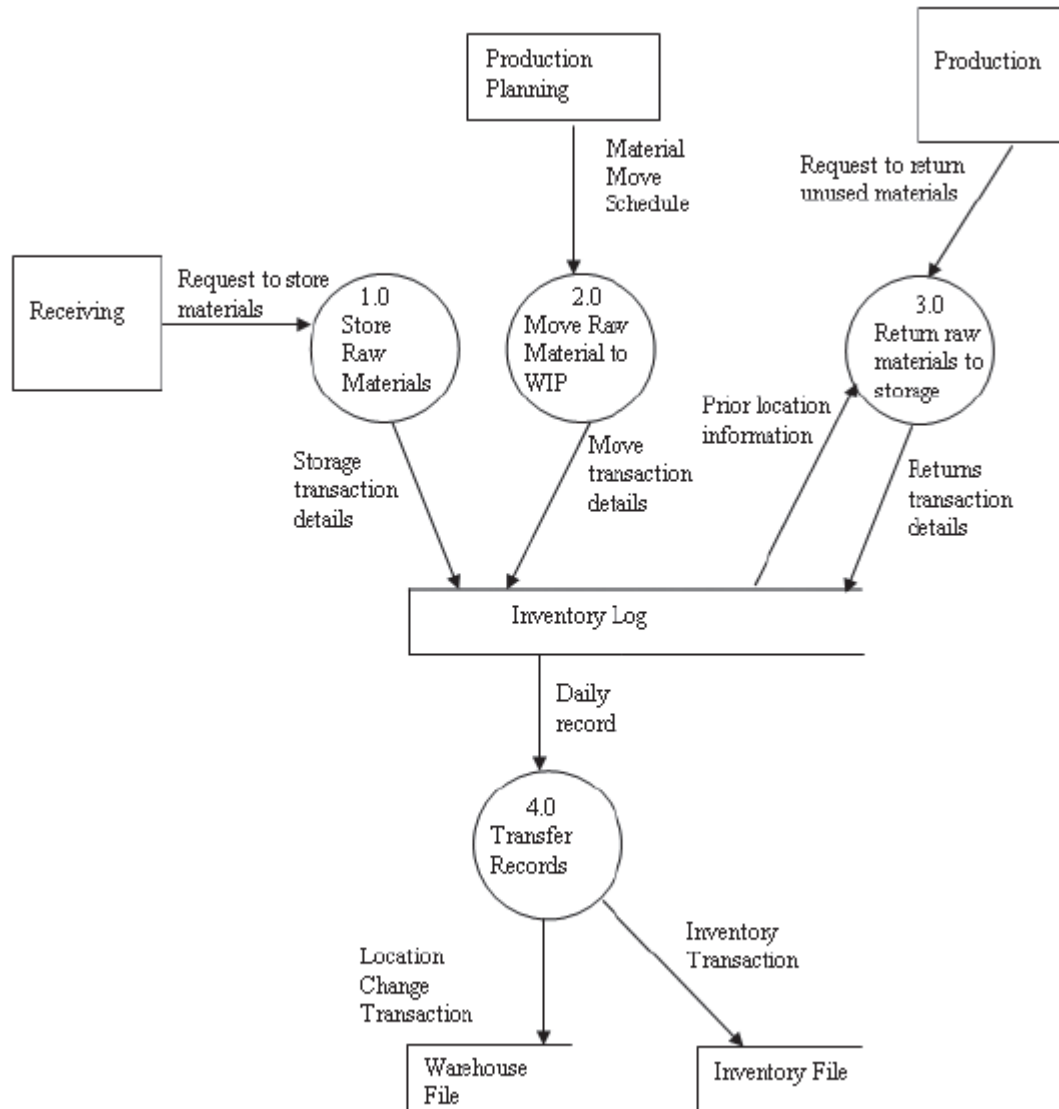


Figure 4.17 Decomposition of control stored materials.

- **Process 1.0** is triggered by the data flow “**Request to Store Materials**” and provides input to the inventory Log by the data flow “**Storage Transaction Details.**” The “**Inventory Log**” is a data store that is internal to the process. It is a temporary data store.
- **1.0 Store raw materials** : **Receiving** makes requests to forklift truck drivers to move material from the loading dock to inventory **storage**.
- **2.0 Move raw material to WIP**: At the beginning of the shift the forklift truck driver is given **the schedule of material moves** from storage to the factory floor.
- **3.0 Return unused raw material to storage**: Some materials that are brought to the factory floor may be **returned** if they are not used.
- **4.0 Transfer records**: The forklift truck driver’s inventory log is used as the primary record for updating the warehouse and inventory records. This updating is done at the end of the shift.